EasiGP

User Manual

Ensemble AnalySis with Interpretable Genomic Prediction
A guide to every GUI field and the configuration flow

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1. About EasiGP

EasiGP runs several genomic prediction models on the same data, combines them into ensembles, and shows what each model has learned at the genome level as a circos plot. Everything is configured in a web-based graphical interface (GUI); no coding is needed for day-to-day use. A run produces prediction-accuracy tables and plots, Excel summary reports, marker effects, marker-pair interactions and circos plots.

What you can do with it

- **Predict traits** with up to 15 individual models (BLUP/Bayesian, tree-based, kernel, neural-network and graph attention models).
- **Combine models** into a naive (equal-weight) ensemble and/or weighted ensembles.
- **Interpret models**: marker effects, marker-pair interactions and attention weights, drawn side by side on circos plots together with known gene regions.
- **Prepare data** (optional): LD pruning, Random-Forest marker filtering, ASReml-R BLUE/BLUP adjustment, and a gene-interaction network for the biological-prior GAT model.
- **Scale up**: run on a laptop or generate HPC job scripts (Slurm/PBS), including job arrays that resume automatically after a failure.
- **Compare finished runs** in Tool mode without refitting anything.

Key terms used in this manual

Term	Meaning
Task (scenario)	One unit of work: a population (or population pair) × phenotype × split ratio × replicate. Every selected model is fitted once per task.
Replicate / iteration	A repeat of the random train/test split with a different random draw. More replicates show how stable the results are.
Split ratio	How individuals are divided into training, (validation) and test sets, e.g. 0.8 or (0.8, 0.1, 0.1).
Validation set	Held-out individuals used to tune hyperparameters or learn ensemble weights. The test set is never used for this.
Marker effect	How strongly a marker contributes to a model’s predictions (coefficients or Shapley values).
Interaction	A score for how strongly a pair of markers acts together in a model.
Ensemble	A combination of the predictions (and effects) of the selected models.
Batch	In Parallel mode, a group of tasks processed by one HPC array sub-job.

2. Installing and launching

There are two ways to install EasiGP. Full step-by-step commands are in the README of the GitHub repository; this table summarises the choice.

Route	Where	What you do	Choose it when
A. Ready-made image (recommended)	Laptop: Docker HPC: Apptainer	Pull `shunichirot/easigp` from Docker Hub and run it; nothing else to install.	You want the simplest setup.
B. From source code	Laptop or HPC (conda)	Clone the repository, create the conda environment from `Environments/*.yml`, install `plink2` if you need LD pruning.	You want to read/modify the code, or use ASReml-R (not included in the images).

Full vs Light version. Both contain all models, tuning, preprocessing and plots. The Full version adds the Claude Code integration used to generate gene-interaction networks with FLASH-P automatically (needs a Claude access token). With the Light version you can still supply a network file yourself.

2.1 Starting the GUI

From the main EasiGP folder (with the environment activated):

```
streamlit run main_app.py --server.port 8501
```

Then open `http://localhost:8501` in a browser. On an HPC, start the app inside an interactive job and open an SSH tunnel from your own computer (see the README), or use a virtual desktop (e.g. UQ Bunya onDemand, NCI ARE). Press **Ctrl+C** in the terminal to stop.

2.2 The interface at a glance

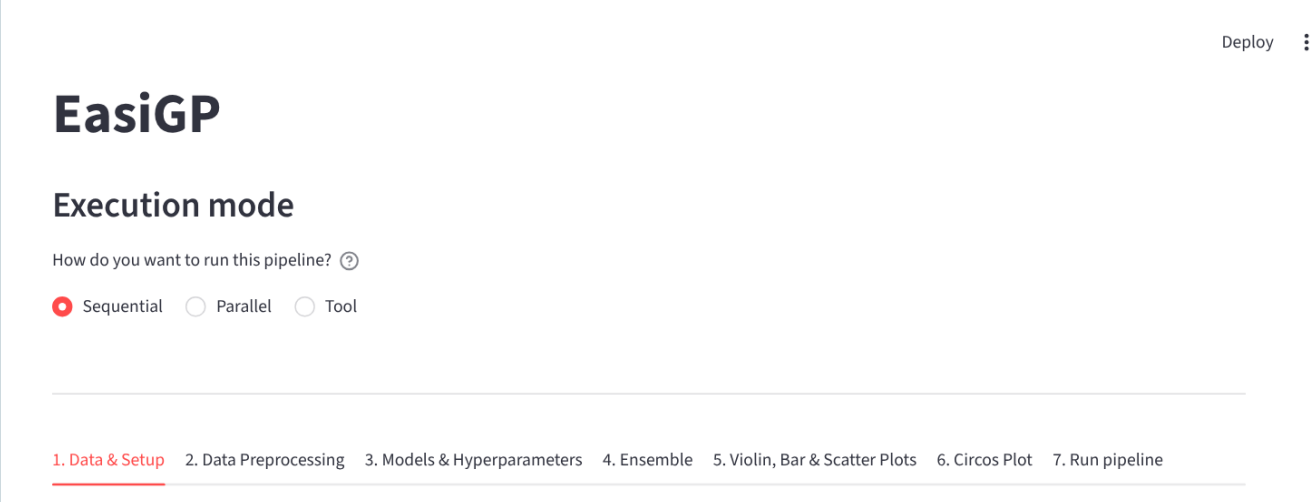


Figure 1. Top of the GUI: the execution mode (Sequential, Parallel or Tool) and the configuration tabs.

- **Execution mode** at the top decides which tabs appear (Section 4).
- **Tabs 1–7** are worked through from left to right (Section 5).
- Every field has a **?** icon: hover over it for a detailed explanation of that field.
- Many plot fields show **auto** or are pre-filled from your data; leave them as they are unless you want to override them.
- **Your settings are remembered**: the GUI saves its state automatically and restores it the next time it starts. Use **Load a previous configuration** (top of Tab 1) to restore the settings of an earlier run from its saved configuration file.

3. Preparing your input data

EasiGP reads several small, plain CSV files. Only the genotype and phenotype files are needed to fit models; the others are needed for circos plots or for specific add-ons.

File	Needed?	Columns (in this order)
Genotype CSV	Yes (or PLINK)	ID, population, then one column per marker. Values are 0/1/2 allele counts.
Genotype PLINK1 fileset	Alternative to CSV	`stem.bed` / `.bim` / `.fam` sharing one file stem; useful for large marker sets.
Phenotype CSV	Yes	ID, population, then one column per trait. In “Raw” layout it may also hold design columns (block, row, year, ...).
marker_info.csv	For circos plots	chromosome, name, start, end. *name* must match the genotype marker columns exactly.
chrom.csv	For circos plots	chromosome, start (usually 0), end (length), optional population.
gene_info.csv	Optional	chromosome, start, end, name, colour (category), source (ring label), phenotype, optional population.
network.json + gene location table	Bio-prior GAT only	Which genes interact, and where each gene is (Gene_Name, Chromosome, Start_bp/End_bp or Start_cM/End_cM).

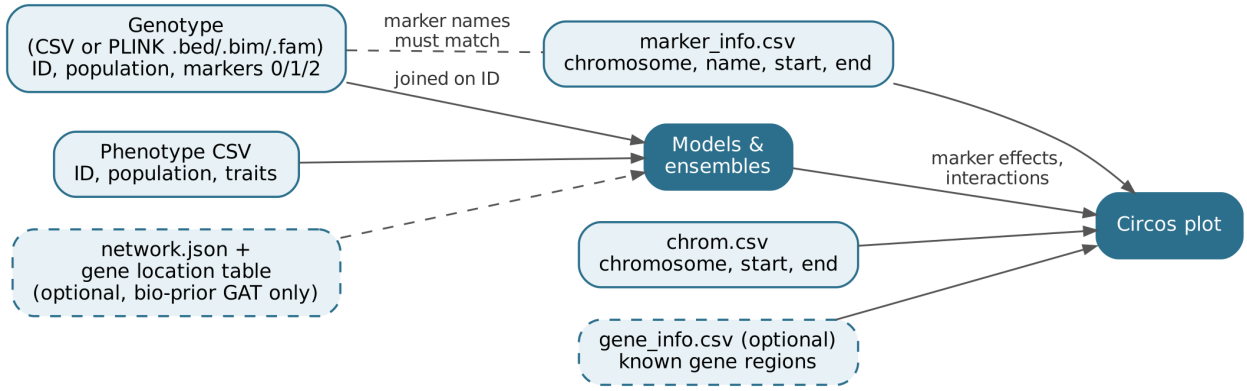


Figure 2. How the input files connect. Dashed boxes are optional.

Rules that avoid most errors

- The **first two columns** of the genotype and phenotype files are read **by position** (ID, population), whatever their header says. All other columns are matched **by name**.
- Marker, chromosome and gene-info files are also read by column position: keep the column order shown above.
- Chromosome names must be identical across `chrom.csv`, `marker_info.csv` and `gene_info.csv` (e.g. *10* everywhere, not *10* and *A10*).
- With PLINK input, ID and population still come from the phenotype file; the .fam IDs are only used for matching.
- Example data: `Data/MaizeNAM`, `Data/TeoNAM` and `Data/Arabidopsis`. The GUI’s default paths point to MaizeNAM population 1, so you can try EasiGP straight away. Unzip any `.zip` files first.

4. Choosing an execution mode

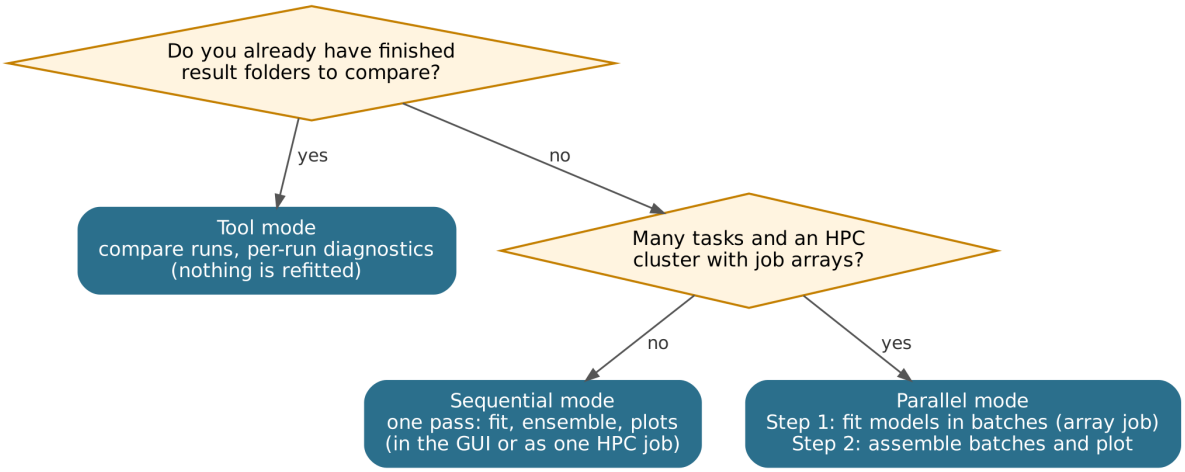


Figure 3. Which execution mode to choose.

Mode	What it does	Tabs shown
Sequential	Runs everything in one pass: data setup, model fitting, ensembles, plots. Run it in the GUI or export it as a single HPC job.	1, 2, 3, 4, 5, 6, 7
Parallel – Step 1	Fits the models, split into batches of tasks that run as independent HPC array sub-jobs.	1, 2, 3, 4, 7
Parallel – Step 2	Assembles the results of all batches and produces every plot and report.	1, 4, 5, 6, 7
Tool	Compares finished runs and runs per-run diagnostics. Nothing is refitted.	Its own two tabs

Tip
Start with Sequential mode on a small dataset (e.g. the MaizeNAM example) to check that your configuration works, then switch to Parallel mode for the full analysis on an HPC. Settings are kept when you switch mode.

5. The configuration flow

Work through the tabs from left to right. Each tab feeds the next, and Tab 7 finally runs the pipeline (or writes an HPC job script).

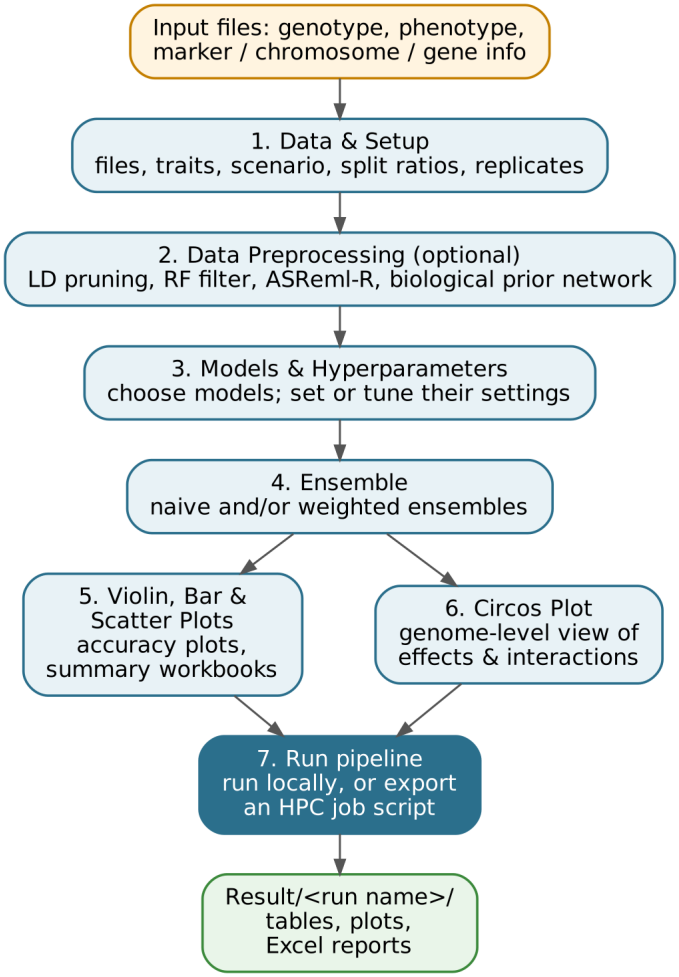


Figure 4. Configuration flow through the tabs (Sequential mode).

5.1 Quick start with the example data

This run uses only default values and takes a few minutes on a laptop.

1. Select **Sequential** as the execution mode.
2. **Tab 1**: keep the MaizeNAM population 1 paths and choose a **Target phenotype** such as *days2anthesis*.
3. **Tab 3**: keep the default models (rrBLUP, BayesB, RF).
4. **Tab 4**: keep **Naive ensemble (equal weight)** ticked.
5. **Tab 6**: keep the default chromosome, marker and gene info paths.
6. **Tab 7**: choose **Option B: run now (local)** and click **Run pipeline now (local)**.
7. Open `Result/MaizeNAM/` to see the tables, plots and circos plots (Section 13).

6. Tab 1 – Data & Setup

Tells EasiGP where the data are, which traits to predict and how individuals are split.

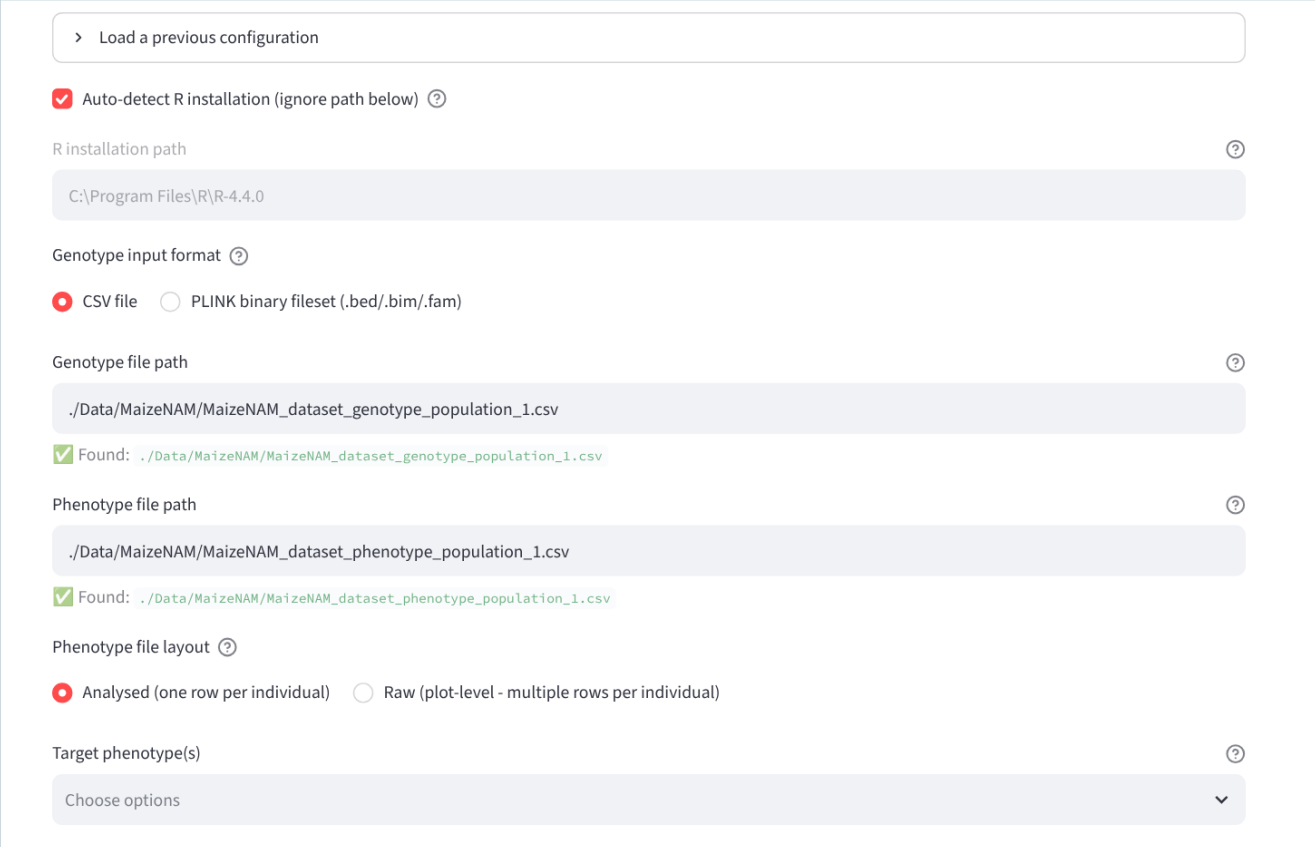


Figure 5. Top of Tab 1 with its default values.

Field	Role
Load a previous configuration	Restores every field from a saved configuration file of an earlier run (pick from the list, type a path, or upload).
Auto-detect R installation / R installation path	R runs rrBLUP, GBLUP, BayesB and RKHS. Keep auto-detect on; untick only if R is not found and enter its folder.
Genotype input format	CSV file, or PLINK binary fileset (enter the file stem without extension and the plink2 executable).
Genotype / Phenotype file path	Paths to your files. A green “Found” message confirms the file exists.
Phenotype file layout	**Analysed**: one row per individual (default). **Raw**: plot-level records, several rows per individual; needed for the ASReml-R stage. In Raw layout you also pick the trait columns and how “population” is derived (from one or several columns, or one constant label).
Target phenotype(s)	Trait column(s) to predict, or *all*. Each trait is analysed separately.
Direction per phenotype	For Top-k Coincidence and Czekanowski: whether the “top” individuals have the highest values (default, e.g. yield) or the lowest (e.g. disease score, days to maturity).
Result folder name	Results are written to `Result/<name>/`. Re-using a name resumes an unfinished run.
Result file compression	*gzip* (default) compresses the large result tables (.csv.gz); *none* writes plain CSV.
Prediction scenario	**within**: train and test within each population. **between**: train on one population, test on another (Figure 6).
Split ratio(s)	within: a training share such as *0.8*, or a (train, validation, test) tuple such as *(0.8, 0.1, 0.1)*. Several values separated by commas are each run separately.

Field	Role
Number of random-split iterations	How many replicates of the random split are run.
Use a validation set / Validation set source	between only: whether to use a validation set and where it comes from (a third population, or part of the training or test population).
Population combination(s) to exclude	between only: remove specific train→test (or train→validation→test) combinations.
Minimum data points required per iteration	Tasks with fewer individuals than this (after removing missing values) are skipped. Default 100.

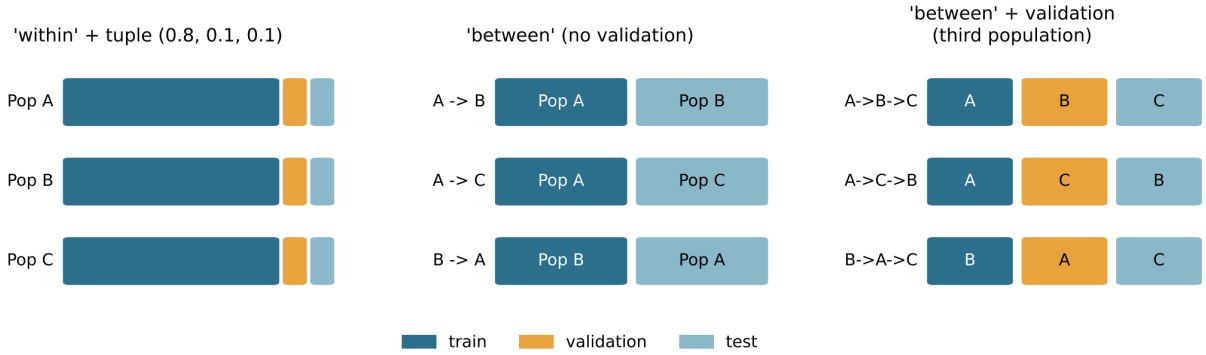


Figure 6. How individuals are assigned in the within and between scenarios.

Important: when you need a validation split
Any **weight-optimisation ensemble** (Tab 4) and any **hyperparameter tuning** (Tab 3) need a validation set. In the *within* scenario, type the split ratio as a tuple, e.g. `(0.8, 0.1, 0.1)`; in the *between* scenario, tick **Use a validation set**. Otherwise a plain number such as `0.8` is enough.

7. Tab 2 – Data Preprocessing (optional)

Four independent sections. All are off by default; switch on only what you need. LD pruning and RF filtering are decided from each task's **training set only**, so no information leaks from the test set. When both are on, LD pruning runs first and RF filtering narrows the remaining markers.

7.1 LD pruning

Removes markers in high linkage disequilibrium using plink2.

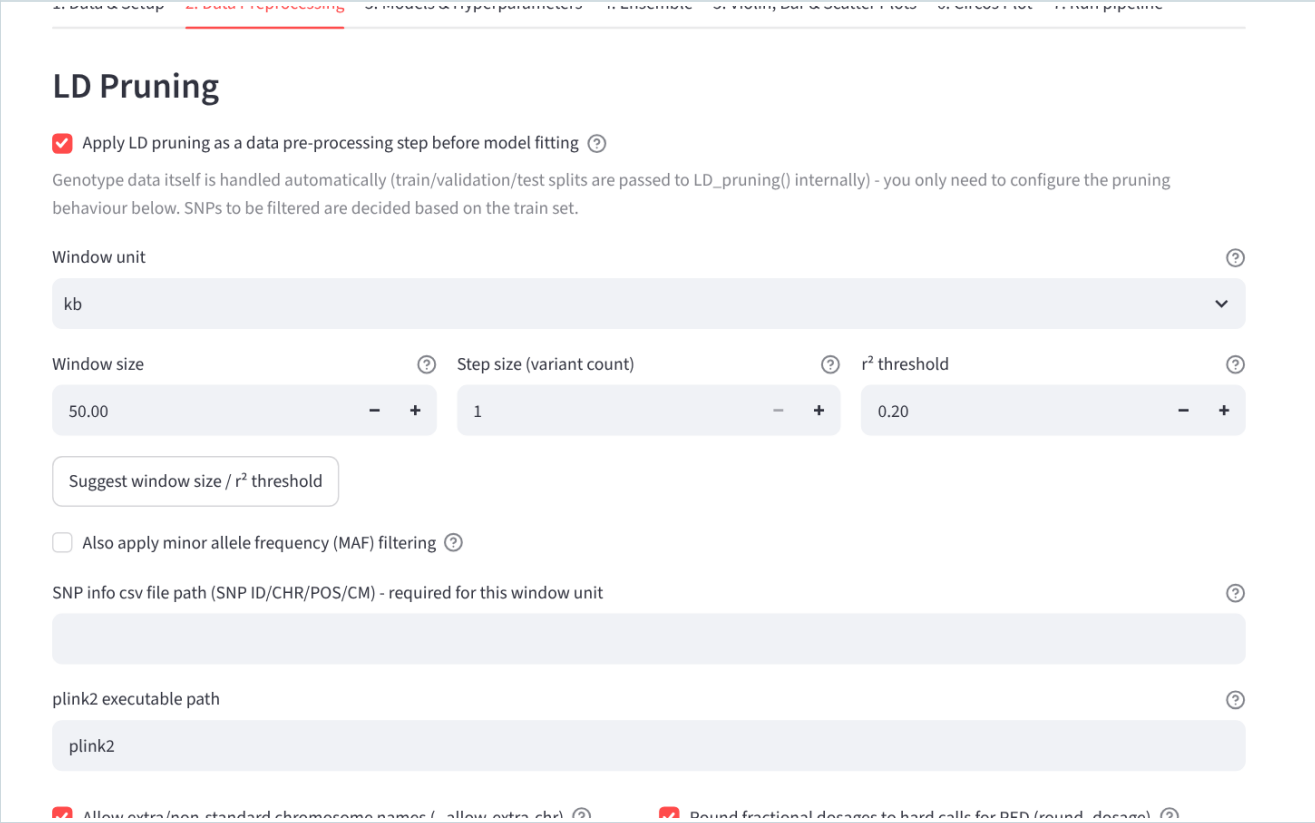


Figure 7. LD pruning settings (shown after ticking the checkbox).

Field	Role
Window unit	*kb*, *variants* (marker count) or *cm*. kb and cM need a SNP info CSV with positions.
Window size / Step size / r^2 threshold	Sliding window, how far it moves, and the r^2 above which one marker of a pair is removed (default 50, 1, 0.2). **Suggest window size / r^2 threshold** proposes values from your data.
MAF filtering	Optionally also removes markers with minor allele frequency below the threshold (default 0.05).
SNP info csv / plink2 executable path	Marker positions (SNP ID, CHR, POS, CM) and the plink2 program.
Non-human genome / Haploid chromosome count	Tell plink2 the number of chromosome pairs of your species (e.g. maize = 10).
Allow extra chromosome names / round dosages	plink2 compatibility options; keep the defaults unless plink2 reports an error.
Working directory / Keep intermediate files	Where plink2 writes temporary files, and whether to keep them for inspection.
Unmapped-SNP strategy	For kb/cM windows: markers without a position are pruned by marker count (*variant_count*), kept (*skip*) or removed (*drop*).
Generate LD decay plots	Optional diagnostic plots of r^2 against distance, per population and replicate, with advanced plot settings.

7.2 RF marker importance filtering

Keeps only the most important markers according to a Random Forest fitted to the training set.

Field	Role
Apply RF-based marker importance filtering	Switches the filter on.
How should the number of markers be specified?	**Percentage of markers** (default top 20 %) or **Fixed number of markers (M)**.
Advanced Random Forest settings	Number of trees (200), max features per split, max depth, min samples per leaf, random seed.
Use GPU-accelerated Random Forest	Experimental; uses a GPU implementation when available.

7.3 ASReml-R mixed-model adjustment (BLUE/BLUP)

Fits a linear mixed model to **raw, plot-level** phenotypes and replaces each individual's phenotype with its BLUE or BLUP before any prediction model sees it. ASReml-R is commercial software with its own paid licence; it is not included in the ready-made images (use installation route B).

Field	Role
Fit an ASReml-R mixed model ...	Switches the stage on. Set **Phenotype file layout = Raw** on Tab 1 first. The section shows a green message once ASReml-R is installed and licensed; otherwise an **Activate ASReml-R licence** box appears.
Genotype column	The column that identifies the individual in the raw file.
Genotype effect	One or more of **BLUEs** (genotype fixed, recommended default), **BLUPs without GRM** (random, independent) and **BLUPs with GRM** (random, with a genomic relationship matrix). Several options are run side by side and compared.
Mask validation/test phenotypes with NA	Fits the mixed model without the validation/test records, so they cannot influence their own adjusted values.
Two-step refined BLUEs / weighting by reliability	Optional refinements of the BLUE option.
Marker source / Shrinkage (GRM options)	Which markers build the relationship matrix and how much it is shrunk towards identity before inversion.
Calculate adjusted prediction accuracy	Adds heritability (classic and Cullis) and adjusted prediction accuracy to `Metric.csv`.
Fixed effects / Random effects	Design columns of the raw file (e.g. Location, Year, Block). Each column may have only one role.
Variance structures, residual structure, interaction/nesting groups	Optional advanced model terms (e.g. spatial row × column residuals).
Advanced ASReml-R settings	Workspace, update rounds, behaviour on non-convergence, strict licence check, warm start, threads.
ASReml-R syntax (editable)	Preview of the generated model. You may hand-edit it, then click **Commit syntax**.

7.4 Biological prior network

Only used by **GAT_biological_prior_knowledge**; select that model on Tab 3 to unlock this section. The model treats each *gene* as a graph node (built from the markers inside it), connected by a gene-interaction network.

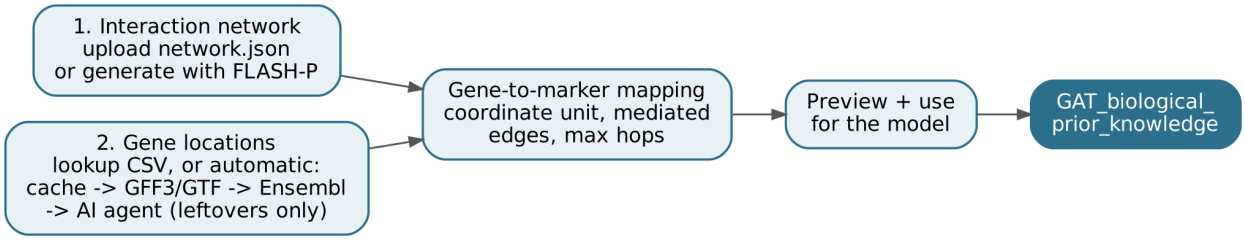


Figure 8. Building the inputs for the biological-prior GAT model.

Field	Role
Species	Passed to FLASH-P and the gene-location lookup.
Coordinate unit / Include mediated edges / Max hops	bp or cM (must match the gene and marker files); whether genes linked through non-gene nodes (e.g. a hormone) are connected, and through how many.
How many networks?	**One network per phenotype**, or **several independent networks** (e.g. repeated FLASH-P runs) to see how much the network changes the result.
1. Interaction network (JSON)	Upload an existing `network.json`, run FLASH-P now (Full version), or locate the output of an earlier FLASH-P run.
2. Gene locations	Supply a lookup CSV, or build it automatically: cached results, a GFF3/GTF annotation file (recommended), Ensembl Plants, then an AI agent for the remaining genes only.
Preview + use for the model	Builds and checks the gene-to-marker mapping and hands it to the model. Unresolved genes are reported and dropped, never guessed.

8. Tab 3 – Models & Hyperparameters

Tick the models to fit. A settings panel opens for every ticked model. All values have sensible defaults; change them only when you have a reason to. XGBoost and EBM need the optional `xgboost` / `interpret` packages and are greyed out if these are missing.

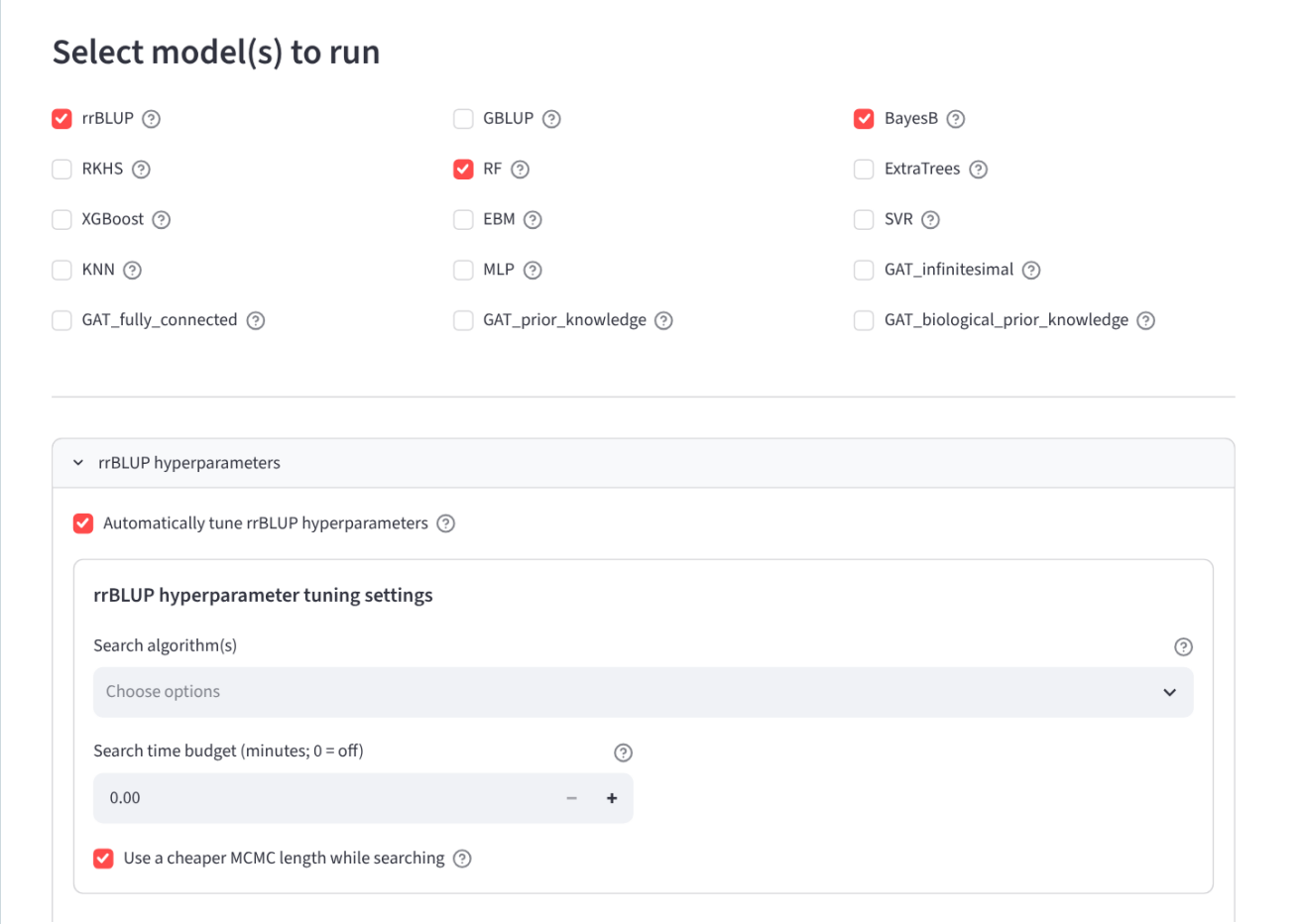


Figure 9. Model selection and the rrBLUP panel with automatic tuning switched on.

8.1 Available models

Model	Family (engine)	Best suited to	Interpretation output
rrBLUP	Linear BLUP (R, BGLR)	Many small additive effects; fast baseline	Effects; approx. interactions
GBLUP	Relationship-based BLUP (R)	Additive genetics via relatedness	Shapley effects; approx. interactions
BayesB	Bayesian sparse (R)	A few large-effect markers	Effects; approx. interactions
RKHS	Kernel (R)	Non-additive, similarity-based	Shapley effects; approx. interactions
RF	Random forest (scikit-learn)	Non-linear effects and interactions	SHAP effects; TreeSHAP interactions
ExtraTrees	Randomised trees (scikit-learn)	Alternative to RF, less overfitting	SHAP effects; TreeSHAP interactions
XGBoost*	Gradient boosting	Fast, flexible non-linear model	SHAP effects; TreeSHAP interactions
EBM*	Explainable boosting (interpret)	Glass-box model with native pairs	Native pairwise terms
SVR	Support vector regression	Non-linear patterns via kernel	Shapley effects; H-statistic

Model	Family (engine)	Best suited to	Interpretation output
KNN	Nearest neighbours	Simple similarity-based prediction	Shapley effects; H-statistic
MLP	Neural network (PyTorch)	Complex patterns, larger data	Shapley effects; NID interactions
GAT_infinitesimal	Graph attention (self-loops)	Additive-style graph baseline	Shapley effects
GAT_fully_connected	Graph attention (all pairs)	Most flexible graph model; slowest	Effects; attention
GAT_prior_knowledge	Graph attention (RF-selected pairs)	Targeted, faster graph model	Effects; interactions; attention
GAT_biological_prior_knowledge	Graph attention over genes	Uses known gene networks (Section 7.4)	Effects; attention

* Optional packages. “Approx.” = interactions of the four BGLR models come from a surrogate tree model and are approximate rankings.

8.2 Fields common to most model panels

Field	Role
Automatically tune <model> hyperparameters	Searches for the best values of the tunable fields on the validation set (score: Pearson r − MSE/phenotype variance), per task. Needs a validation split.
Search algorithm(s)	**Grid**, **Random**, **Bayesian**, **Nelder-Mead**, **Powell**. Choosing several gives separately tuned variants (e.g. *RF__Grid*, *RF__Bayesian*).
Iterations / Grid points / Search time budget	How many candidates are tried, and an optional time limit.
Model-specific fields	e.g. Iteration number and Burn-in (BGLR models, default 12000 / 2000); Tree number (100) for RF; Neuron numbers, Learning rate, Epoch, Batch size (MLP, GAT).
Return marker effect?	Computes marker effects (for the circos rings). Shapley-based effects can be slow; “Number of samples” controls the cost.
Marker effect method	How effects are computed (e.g. closed form for rrBLUP/BayesB, SHAP for tree models).
Return marker-pair interactions?	Computes interaction scores for the circos links.
Max markers considered for interaction search	Restricts the search to the top markers (default 500) to keep it tractable.
Output only the top N% of interactions	**Highlighted in the GUI**: strongly affects both run time and how interpretable the interactions are. *all* keeps everything.
Marker interaction method	e.g. pairwise SHAP, Friedman H-statistic or shapiq variants, depending on the model.

Global settings at the bottom of Tab 3
When tuning is on, additional shared options appear (Bayesian batch search, warm-start across replicates, **Fast tuning** preset, re-use of tuned values from a pilot run). The **Acceleration settings** expander (shown with the biological-prior GAT) only changes speed, never results. **Fail fast on missing optional model dependencies** stops a run at the start if a selected model’s package is missing.

9. Tab 4 – Ensemble

Choose one or more ways to combine the selected models. Selecting several lets you compare them in the same run.

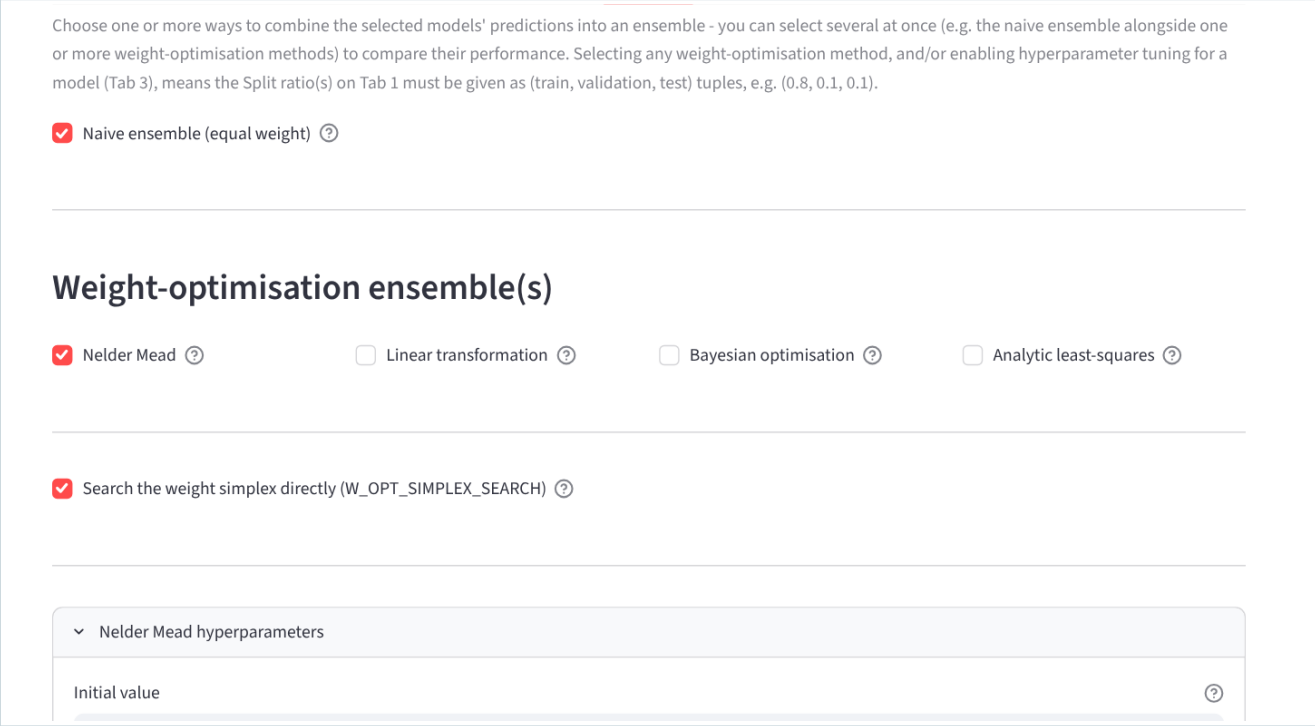


Figure 10. Ensemble choices with Nelder Mead selected.

Option	How weights are found	Notes
Naive ensemble (equal weight)	Simple average of all selected models.	Default; needs no validation set.
Nelder Mead	Gradient-free numerical search.	Simple and generally reliable.
Linear transformation	A small neural network learns the weights (trained several times).	Settings: learning rate, epochs, decay, batch size, patience, number of MLP models.
Bayesian optimisation	Uses past attempts to choose the next weights to try.	Can need fewer attempts; more overhead each.
Analytic least-squares	Exact closed-form least-squares solution.	Milliseconds, nothing to tune, no randomness.

Shared field	Role
Search the weight simplex directly	Keeps weights non-negative and summing to 1 during the search (default on).
Minimum / Maximum boundary	Range each model's weight may take (0 lets a weak model be dropped).
Objective	What the search minimises on the validation set (default: ensemble MSE).
Diversity penalty (lambda) / method	Optional pull towards equal weights; 0 switches it off.
Naive-shrinkage alpha	Blends the result towards the naive ensemble: 1 = as found, 0 = naive ensemble. A built-in guard keeps the result no worse than naive on the validation set.
Combine per tuning method / across methods	Appears when a model is tuned with more than one algorithm: one ensemble per algorithm, or one combined ensemble.

Remember
Every weight-optimisation method needs a validation split (Section 6).

10. Tab 5 – Violin, Bar & Scatter Plots

Controls the accuracy plots and summary reports created after fitting. Each section has its own **Create …** checkbox plus Font size, Figure size and Plot quality (DPI). Advanced style expanders are optional.

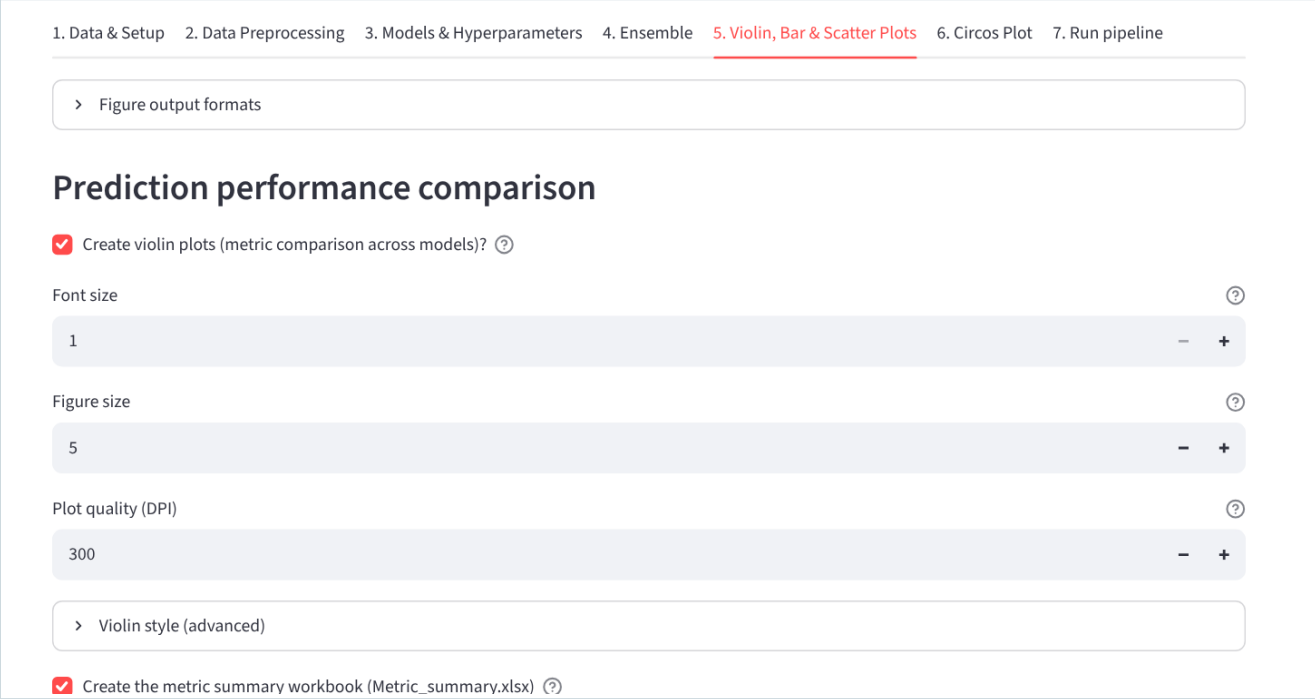


Figure 11. Top of Tab 5.

Section	Output	Notes
Figure output formats	Extra SVG and/or PDF copies of every figure	PNG files are always written.
Prediction performance comparison	Violin plots of Pearson correlation and MSE per model; `Metric_summary.xlsx`	Optional calibration-slope (“dispersion”) sheet.
Diversity Prediction Theorem	`Diversity_prediction_theorem.xlsx`; ensemble weight plots (`Weight.png`)	Explains ensemble error as average model error minus model diversity. Optional marker/pair stability summary.
Model pairwise comparison	Scatter-plot matrix comparing models’ predictions and marker effects	Optional QTL file to highlight known QTL; colour by population; normalise effects.
Adjusted prediction accuracy	Violin plots	Needs the ASReml-R adjusted accuracy option (Tab 2).
Spearman correlation	Violin plots	Rank-based accuracy.
Top-k Coincidence	Curves	Share of the true top-k% individuals that a model also ranks in its top-k%. Uses the direction set on Tab 1.
Czekanowski coefficient	Pairwise model matrices	Agreement between two models’ top-k% selections; choose thresholds.
Pairwise comparison of metrics	Pearson vs Spearman (and adjusted) scatter plots	
Comparison between BLUEs / BLUPs	Each metric compared across ASReml-R options	Needs two or more Genotype-effect options.

11. Tab 6 – Circos Plot

A circos plot shows, for each phenotype and population, the genome as a circle with one ring of marker effects per model, known gene regions, and links between interacting marker pairs. It lets you compare the genetic architecture inferred by different models and spot regions not yet described in the literature.

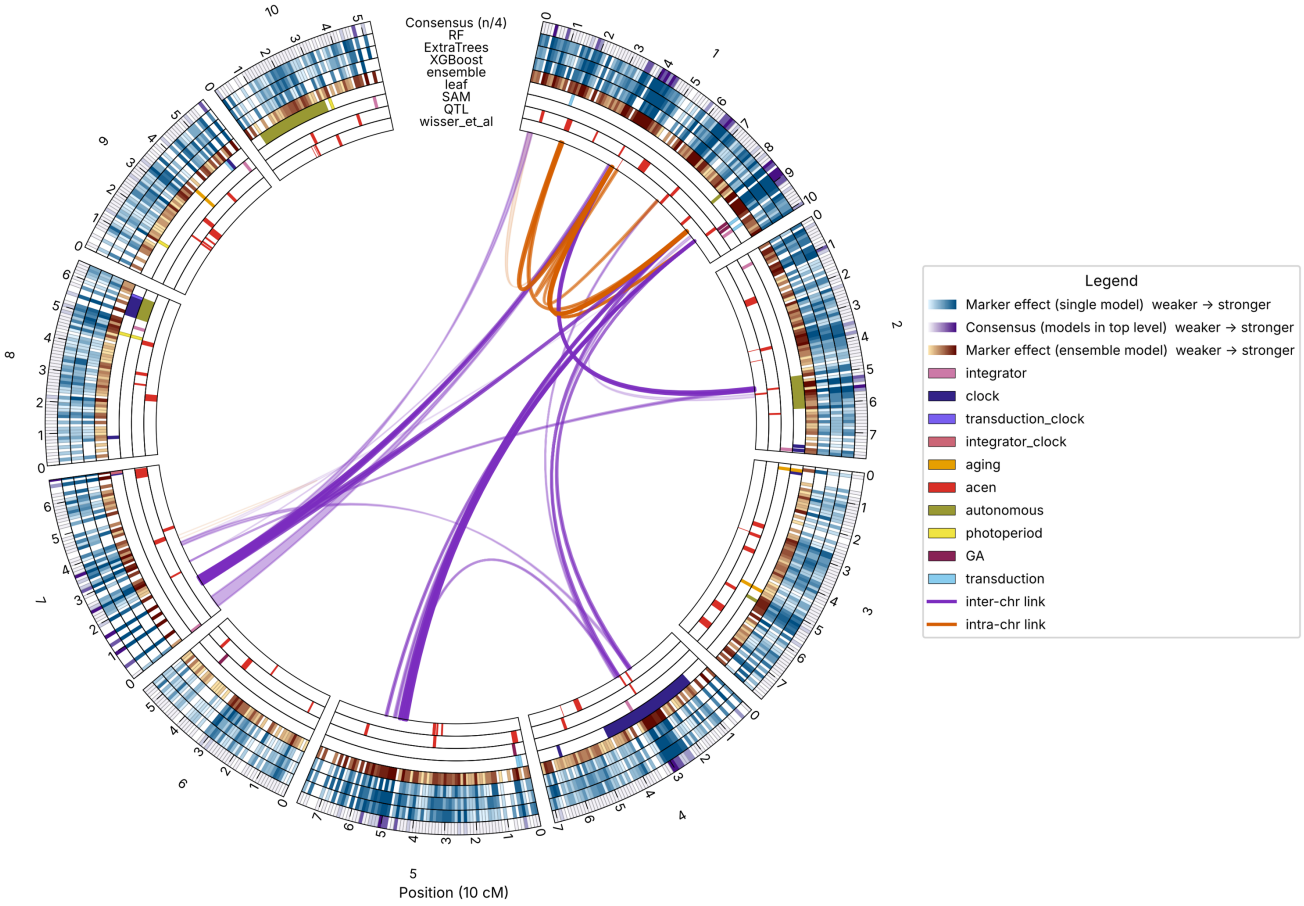


Figure 12. A circos plot produced by EasiGP for the MaizeNAM example (population 1, days2anthesis; RF, ExtraTrees, XGBoost and the naive ensemble, 3 replicates) with the shared legend. Rings from outside in follow the label list at the top: chromosome ticks (10 cM units), the consensus ring (purple), one marker-effect ring per model (blue), the ensemble (brown), then the known gene-region rings from gene_info.csv coloured by category. Links in the centre are the 30 strongest marker-pair interactions of the ensemble (purple between chromosomes, orange within a chromosome).

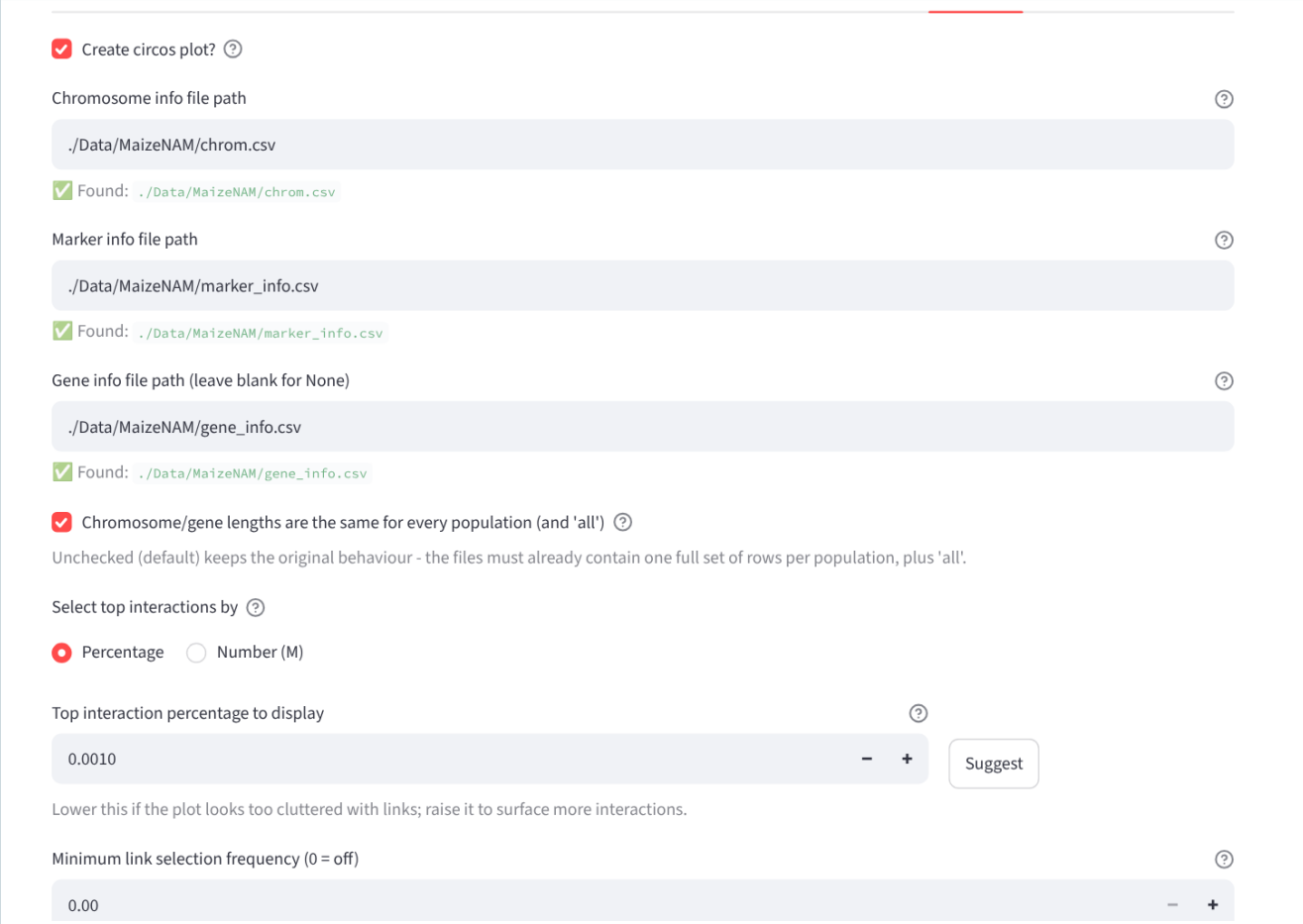


Figure 13. Top of Tab 6; layout values are auto-filled from the MaizeNAM example files.

Field	Role
Create circos plot?	Switches circos plotting on or off.
Chromosome / Marker / Gene info file path	The `chrom.csv`, `marker_info.csv` and optional `gene_info.csv` (Section 3).
Chromosome/gene lengths are the same for every population	Tick if one set of rows applies to all populations (no population column needed).
Select top interactions by	Percentage or Number (M) of the strongest interactions drawn as links. **Suggest** proposes a value. Lower it if the plot is cluttered.
Minimum link selection frequency	Only draw links selected in at least this share of replicates (0 = off).
Chromosome coordinate unit	bp or cM.
Space between rings, Start/End angle, Scale, Label sizes	Layout. Auto-filled from your files; type a value to override.
Minimum link opacity	How faint the weakest drawn link may be (the strongest is always opaque).
Averaging window size (WINDOW)	Averages marker effects within windows; 0 plots each marker's own effect.
Gene location adjustment (GENE_ADJUST)	Widens gene regions so they are visible; does not change their true position.
Links, consensus ring and marker labels	Link order, scale and colours; an outer **consensus ring** summarising models; labels for the top N markers.
Highlight interactions from genome regions	Emphasise links touching chosen genes, intervals or chromosomes and fade the rest.
Advanced settings / Edit cytoband colour map	Effect ordering, ring-label fitting, figure size, higher-order interactions ($k \geq 3$); colours for each gene category in `gene_info.csv`.

12. Tab 7 – Run pipeline

First choose a workflow; only the settings for that workflow are shown.

Option	What happens	Use it for
Option A: export config, then submit a job	Writes a configuration JSON and a job script (Slurm or PBS). Nothing runs in the GUI.	Real analyses on an HPC.
Option B: run now (local)	Runs immediately on the machine running the GUI.	Small runs and tests.

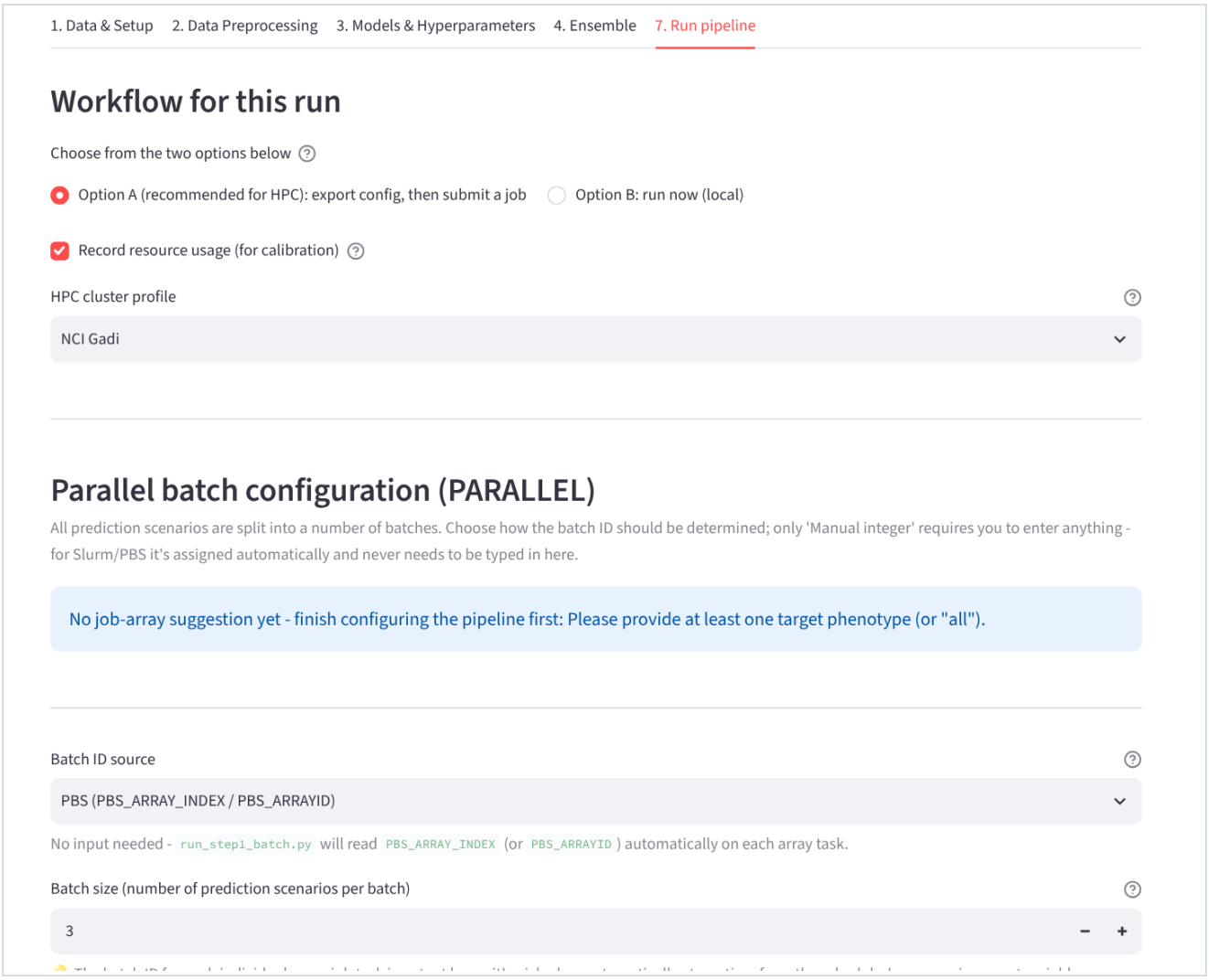


Figure 14. Tab 7 in Parallel / Step 1 with Option A. Suggestion panels fill in once a target phenotype is set.

12.1 Option A fields

Field	Role
Record resource usage (for calibration)	Saves time and memory per task so later resource suggestions are calibrated to your runs. Results are unaffected.
HPC cluster profile	NCI Gadi, UQ Bunya, Generic / other HPC, or Custom (enter limits such as cores and memory per node).
Suggested job-array size / compute resources	Proposed array layout and CPU/GPU/memory/walltime for your configuration. Treat as a starting point.
Batch ID source (Parallel Step 1)	Normally read automatically from the scheduler (SLURM_ARRAY_TASK_ID or PBS_ARRAY_INDEX).
Batch size	Number of tasks per array sub-job (default 3). Larger = fewer, longer sub-jobs.

Field	Role
HPC job resource requests	Scheduler, job name, nodes, CPUs per task, memory, walltime, partition/queue, account, GPU, QoS, array start/end index, max concurrent tasks.
Additional setup lines	Lines added to the script before EasiGP runs, e.g. `module load anaconda3` and `conda activate EasiGP`.
Also write a pilot job script / Pilot tasks (K)	Runs K representative tasks first to measure time and memory before a large submission.
Generate and save job files	Writes the files and shows their location. Submit with e.g. `sbatch <job name>.sh` (Slurm) or `qsub <job name>.sh` (PBS).

12.2 Parallel mode in practice

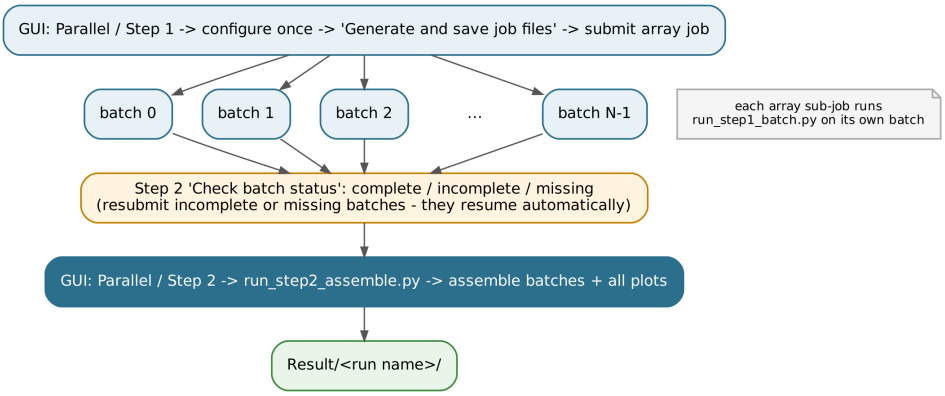


Figure 15. Parallel workflow: Step 1 fits batches as an array job; Step 2 assembles them and plots.

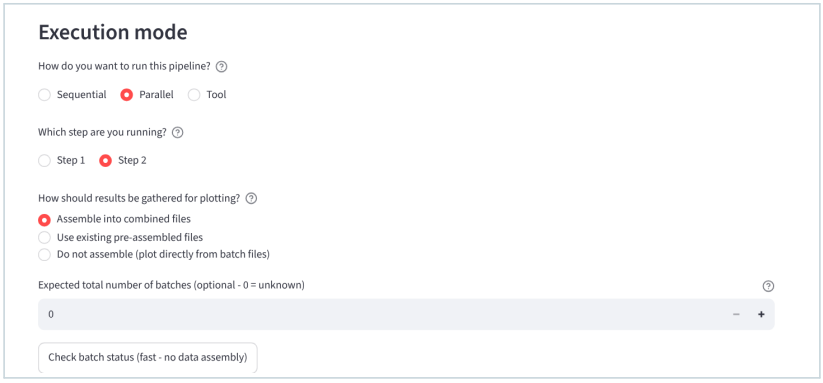


Figure 16. Parallel / Step 2 options at the top of the page.

1. Choose **Parallel → Step 1**, configure Tabs 1–4, then in Tab 7 generate the job files and submit the array job.
2. When the jobs finish, choose **Parallel → Step 2** with the **same Result folder name**. Click **Check batch status** to list complete, incomplete and missing batches.
3. Resubmit incomplete or missing batches with the same configuration: finished tasks are kept and only unfinished ones are re-run.
4. Configure Tabs 5–6 and run Step 2. **How should results be gathered**: *Assemble into combined files* (normal), *Use existing pre-assembled files* (only re-plot), or *Do not assemble* (plot straight from batch files).

13. Tool mode – comparing finished runs

Tool mode reads the saved result files of finished runs and never refits a model. It has two tabs.

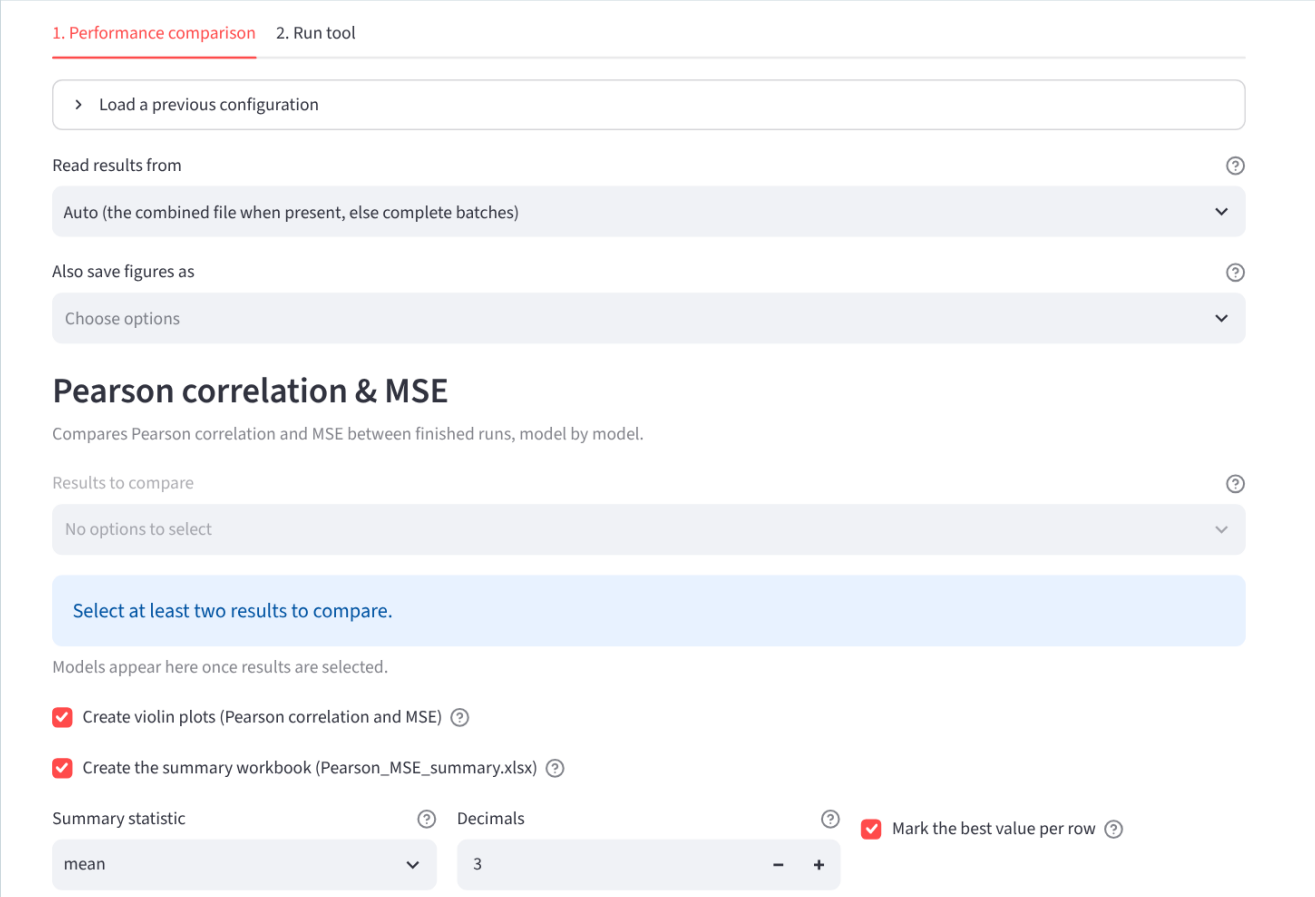


Figure 17. Tool mode, tab 1 (Performance comparison).

Section	Role
Read results from	Auto, combined (assembled) files only, or complete batch files only.
Pearson correlation & MSE / Adjusted accuracy / Spearman	Pick two or more result folders, edit the names shown in figures, choose models; creates violin plots and a summary workbook (statistic, decimals, best value marked).
Comparison of metrics between results	Scatter plots of each metric between pairs of results.
Per-run diagnostics (optional)	Prediction diagnostics (calibration, within-group accuracy), member importance and Shapley value, consensus stability, candidate reliability (needs the genotype file), class hierarchy and pairwise diversity.
2. Run tool	Output folder name inside Result/ (must not be a finished run), then Option A (export job) or Option B (run now).

14. Output files

Every run writes into `Result/<run name>/`. Parallel Step 1 writes one file per batch (e.g. `Metric_3.csv`), which Step 2 merges. Large tables may end in `.csv.gz`.

File	Contents
Metric.csv	One row per model, phenotype, population, ratio and replicate: Pearson correlation, MSE, Spearman, Top-k Coincidence (+ heritability and adjusted accuracy with ASReml-R).
Prediction_result_train / valid / test.csv	Observed and predicted values for each individual.
Marker_effect.csv	Each model's marker effects (circos effect rings).

File	Contents
Interaction.csv / Attention.csv	Marker-pair interaction scores; GAT attention weights.
Weight.csv	Weight of each model in each weighted ensemble.
hyperparameter.csv	Hyperparameters used or found by tuning.
Basic_stats.csv	Split sizes and number of markers per task.
Metric_summary.xlsx / Diversity_prediction_theorem.xlsx	Excel summary reports (plain CSV if openpyxl is missing).
circos_<phenotype>_<population>*.png, circos_legend.png	Circos plots with one shared legend.
Pearson correlation.png, MSE.png, Weight.png, Scatter_plot_*.png, ...	Violin, weight and scatter plots (+ SVG/PDF if requested).
***_config.json**	The configuration used, for repeating the run or reloading it in the GUI.

Do not delete a run folder between resubmissions
Hidden `.checkpoint_*.json` files and partial results let a resubmitted job continue where it stopped.

15. Troubleshooting and tips

Symptom	Likely cause and fix
Run blocked: a validation split is needed	Weight optimisation or tuning is on. Use a tuple split such as (0.8, 0.1, 0.1), or tick **Use a validation set** (between).
Model checkbox greyed out (XGBoost/EBM)	Install the optional package (`pip install xgboost` / `interpret`) and restart the GUI.
R models fail to start	Untick **Auto-detect R installation** and enter the R folder, or activate the correct environment.
Tasks are skipped	Fewer individuals than **Minimum data points required per iteration**; lower it or merge populations.
A gene does not appear on the circos plot	Chromosome names differ between files, or the gene's phenotype column does not match the trait name.
Circos plot is cluttered	Lower **Top interaction percentage** or use **Number (M)**; raise **Minimum link opacity**.
LD pruning fails	Check the plink2 path and set the haploid chromosome count for your species.
"EasiGP code changed since this window started"	The code was updated while the GUI was open; restart the GUI.
HPC job cannot find the environment	Add `module load ...` and `conda activate EasiGP` to **Additional setup lines**.
Some batches incomplete or missing	Use **Check batch status** in Step 2 and resubmit the listed batch IDs; they resume automatically.
Run is very slow	Reduce interaction search (top N%, max markers), Shapley sample numbers, number of replicates or models; use Parallel mode on an HPC.

Configuration checklist

Step	Check
Data	Files found (green messages); target phenotype(s) selected; result folder named.
Split	Tuple ratio or validation set if tuning or weighted ensembles are used.
Preprocessing	Only the sections you need are ticked; ASReml-R licensed if used.
Models	Models ticked; interaction settings reasonable for run time.
Ensemble	Naive and/or weighted methods selected.
Plots	Circos files set; plot sections ticked.
Run	Option B for a test; Option A with setup lines for HPC.

Citation. If you use EasiGP, please cite: Tomura S. et al. Ensemble AnalySis with Interpretable Genomic Prediction (EasiGP): computational tool for interpreting ensembles of genomic prediction models. *The Plant Genome*. doi:10.1002/tpg2.70138.